

$$\text{VAR}(\mathbf{X}) = \frac{1}{n-1} \sum_{i=1}^n (x_i - E(X))^2 \quad (1)$$

$$\text{COV}(\mathbf{X}, \mathbf{Y}) = \frac{1}{n-1} \sum_{i=1}^n (x_i - E(X))(y_i - E(Y)) \quad (2)$$

$$\text{COR}(\mathbf{X}, \mathbf{Y}) = \frac{\text{COV}(\mathbf{X}, \mathbf{Y})}{\sqrt{\text{VAR}(\mathbf{X})\text{VAR}(\mathbf{Y})}} \quad (3)$$

$$\mathbf{R}^2 = 1 - \frac{\text{VAR}(\mathbf{X}, \mathbf{Y})_{\text{FittedLine}}}{\text{VAR}(\mathbf{X}, \mathbf{Y})_{\text{Mean}}} \quad (4)$$